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FIRST-ORDER MARKOV CHAIN
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=== Train/Test Split ===
Total ActivityIDs: 62
Train ActivityIDs: 49 (79%)
Test ActivityIDs: 13 (21%)
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```
Simulations = 1000
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TRANSITION MATRICES
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=== Transition Matrices ===
=== Function/Formula Explanation ===
```

Calculate first-order transition probability matrices using:
 $P(x_j | x_i) = x_{ij} / \sum_{m=1 \text{ to } M} x_{im}$

Parameters:
 data: DataFrame with transition data
 from_col: Column name for current state
 to_col: Column name for next state
 categories: List of category names

Returns:
 - First-order transition probability matrix

```
=== Observed 5x5 Matrix (from training data) ===
```

Next_Category	Equipment	Patient	NearPatient	FarPatient	HygieneArea
SurfaceCategories					
Equipment	0.576	0.204	0.033	0.100	0.086
Patient	0.312	0.520	0.040	0.075	0.052
NearPatient	0.216	0.108	0.405	0.162	0.108
FarPatient	0.210	0.134	0.038	0.522	0.096
HygieneArea	0.137	0.021	0.000	0.185	0.658

```
=== Observed 7x7 Matrix (from training data) ===
```

Next_Category	In	Equipment	Patient	NearPatient	FarPatient
HygieneArea	Out				
SurfaceCategories					
In	0.00	0.020	0.000	0.000	0.980
0.000 0.000					
Equipment	0.00	0.572	0.203	0.033	0.100
0.085 0.007					
Patient	0.00	0.312	0.520	0.040	0.075
0.052 0.000					

NearPatient	0.00	0.216	0.108	0.405	0.162
0.108 0.000					
FarPatient	0.00	0.165	0.105	0.030	0.410
0.075 0.215					
HygieneArea	0.07	0.127	0.019	0.000	0.171
0.608 0.006					
Out	0.00	0.000	0.000	0.000	0.000
0.000 0.000					

=== Starting State Probabilities (from training data) ===

5x5:

SurfaceCategories

HygieneArea 1.0

Name: proportion, dtype: float64

7x7:

SurfaceCategories

In 0.776

HygieneArea 0.224

Name: proportion, dtype: float64

=== Test 5x5 Matrix (from unseen data) ===

Next_Category	Equipment	Patient	NearPatient	FarPatient	HygieneArea
SurfaceCategories					
Equipment	0.576	0.174	0.054	0.130	0.065
Patient	0.500	0.400	0.000	0.100	0.000
NearPatient	0.778	0.000	0.111	0.000	0.111
FarPatient	0.552	0.034	0.069	0.310	0.034
HygieneArea	0.038	0.038	0.038	0.192	0.692

=== Test 7x7 Matrix (from unseen data) ===

Next_Category	In	Equipment	Patient	NearPatient	FarPatient
HygieneArea	Out				
SurfaceCategories					
In	0.000	0.000	0.000	0.000	1.000
0.000 0.00					
Equipment	0.000	0.576	0.174	0.054	0.130
0.065 0.00					
Patient	0.000	0.500	0.400	0.000	0.100
0.000 0.00					
NearPatient	0.000	0.778	0.000	0.111	0.000
0.111 0.00					
FarPatient	0.000	0.381	0.024	0.048	0.214
0.024 0.31					
HygieneArea	0.071	0.036	0.036	0.036	0.179
0.643 0.00					
Out	0.000	0.000	0.000	0.000	0.000
0.000 0.00					

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CONVERGENCE

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=== Convergence Analysis ===

Tracking transition (Main): Equipment → Patient

Tracking transition (All): In → Equipment

Main Categories Convergence:

9 episodes: P = 0.2432 (9/ 191 transitions)

18 episodes: P = 0.2987 (23/ 326 transitions)

27 episodes: P = 0.2613 (29/ 455 transitions)

36 episodes: P = 0.2289 (46/ 645 transitions)

45 episodes: P = 0.2259 (54/ 737 transitions)

49 episodes: P = 0.2045 (55/ 782 transitions)

All Categories Convergence:

9 episodes: P = 0.1111 (1/ 212 transitions)

18 episodes: P = 0.0556 (1/ 367 transitions)

27 episodes: P = 0.0370 (1/ 516 transitions)

36 episodes: P = 0.0278 (1/ 727 transitions)

45 episodes: P = 0.0222 (1/ 835 transitions)

49 episodes: P = 0.0204 (1/ 888 transitions)

=== Convergence Assessment ===

Main categories (Equipment → Patient): UNSTABLE ($\sigma = 0.0109$)

All categories (In → Equipment): STABLE ($\sigma = 0.0031$)

Minimum episodes for stability:

Main categories: ~49 episodes

All categories: ~45 episodes

Current dataset: 49 episodes

Is current data sufficient?

Main categories: YES

All categories: YES

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SIMULATIONS

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=== Simulations ===

Running 1000 simulations for each method...

Average Care Period Length: 19

5x5 Fixed Length Simulations: 0.36 seconds

7x7 Terminating State Simulations: 0.42 seconds

5x5 Fitted Length Simulations: 0.36 seconds

=== Sequence Length Distribution Fitting ===

Observed lengths: min=4, max=45

Mean=19.12, Variance=73.58
Variance/Mean ratio: 3.85 (>1 suggests over-dispersion)

Distribution comparison (AIC = Akaike Information Criterion, lower is better):
Poisson($\lambda=19.12$): AIC = 418.84
NegBinom($n=6.72$, $p=0.260$): AIC = 348.76 <-- BEST FIT

Goodness of fit test (Kolmogorov-Smirnov):
D-statistic: 0.0998
P-value: 0.6764
Result: Good fit ($\alpha=0.05$)

=== Simulations with Fitted Length Distribution ===
Using: NegBinom($n=6.72$, $p=0.260$)
Sampled lengths: min=3, max=53, mean=19.18

Fitted-length simulations completed in 0.36 seconds

=== Predicted 5x5 Matrix (from fixed-length simulations) ===

Next Current	Equipment	Patient	NearPatient	FarPatient	HygieneArea
Equipment	0.576	0.210	0.030	0.097	0.086
Patient	0.311	0.517	0.045	0.071	0.055
NearPatient	0.218	0.120	0.405	0.173	0.084
FarPatient	0.200	0.145	0.037	0.526	0.092
HygieneArea	0.132	0.021	0.000	0.177	0.670

=== Predicted 7x7 Matrix (from terminating-state simulations) ===

Next Out Current	In	Equipment	Patient	NearPatient	FarPatient	HygieneArea
In	0.000	0.022	0.000	0.000	0.978	0.000
Equipment	0.000	0.578	0.191	0.036	0.098	0.089
Patient	0.000	0.311	0.518	0.041	0.081	0.048
NearPatient	0.000	0.217	0.108	0.412	0.150	0.114
FarPatient	0.000	0.177	0.098	0.028	0.412	0.081
HygieneArea	0.081	0.127	0.015	0.000	0.171	0.603
Out	0.000	0.000	0.000	0.000	0.000	0.000

=== Predicted 5x5 Matrix (from fitted-length simulations) ===

Next Current	Equipment	Patient	NearPatient	FarPatient	HygieneArea
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Equipment	0.574	0.210	0.033	0.098	0.085
Patient	0.315	0.517	0.036	0.079	0.052
NearPatient	0.213	0.106	0.419	0.148	0.113
FarPatient	0.206	0.146	0.038	0.514	0.095
HygieneArea	0.139	0.022	0.000	0.185	0.654

5x5 Simulation (Fixed Length)

```
['HygieneArea', 'HygieneArea', 'HygieneArea', 'Equipment', 'Equipment',
'Patient', 'Patient', 'Patient', 'Patient', 'Equipment', 'Patient',
'FarPatient', 'FarPatient', 'FarPatient', 'FarPatient', 'HygieneArea',
'FarPatient', 'FarPatient', 'FarPatient']
```

7x7 Simulation

```
['In', 'FarPatient', 'FarPatient', 'Out']
```

Length: 4

Started with: In

Ended with: Out

5x5 Simulation (Fitted Distribution)

```
['HygieneArea', 'HygieneArea', 'FarPatient', 'Equipment', 'Patient',
'Equipment', 'HygieneArea', 'FarPatient', 'Equipment', 'Equipment']
```

Length: 10

Started with: HygieneArea

Ended with: Equipment

=== Comparison of All Three Simulation Approaches ===

- (a) Fixed length MAE (5x5): 0.0054
- (b) Terminating state MAE (7x7): 0.0027
- (c) Fitted length MAE (5x5): 0.0039

Note: Comparing 5x5 vs 7x7 is not strictly like-for-like because 7x7 has more parameters (49 vs 25) and includes terminating states (In/Out). However, the 7x7 MAE of 0.0027 shows how well the terminating model performs overall.

=== Dictionary Inventory ===

Method	Chains	Avg. Chain Len	MAE
5x5 fixed length	1000	19.00	0.0054
5x5 fitted length	1000	19.18	0.0039
7x7 terminating state	1000	21.06	0.0027

=== Difference between Observed and Predicted Matrices ===

For 5x5 fixed-length matrix

Next_Category	Equipment	Patient	NearPatient	FarPatient	HygieneArea
Equipment	0.000	-0.006	0.003	0.003	-0.000
Patient	0.001	0.003	-0.004	0.004	-0.003
NearPatient	-0.002	-0.012	0.000	-0.011	0.024
FarPatient	0.010	-0.011	0.001	-0.004	0.003
HygieneArea	0.005	-0.001	0.000	0.008	-0.013

For 7x7 terminating-state matrix

Next_Category	In	Equipment	Patient	NearPatient	FarPatient
HygieneArea	Out				
SurfaceCategories					

In	0.000	-0.002	0.000	0.000	0.002
0.000 0.000					
Equipment	0.000	-0.006	0.012	-0.003	0.001
-0.004 -0.000					
Patient	0.000	0.001	0.002	-0.001	-0.006
0.004 0.000					
NearPatient	0.000	-0.001	0.000	-0.006	0.013
-0.006 0.000					
FarPatient	0.000	-0.012	0.007	0.002	-0.002
-0.006 0.011					
HygieneArea	-0.011	-0.000	0.004	0.000	-0.000
0.005 0.003					
Out	0.000	0.000	0.000	0.000	0.000
0.000 0.000					

Mean Absolute Error for 5x5 fixed-length: 0.0054

Mean Absolute Error for 7x7 terminating-state: 0.0027

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COMPARISON METRICS

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==== Difference between Observed and Test Matrices ===

For 5x5 matrix

Next_Category	Equipment	Patient	NearPatient	FarPatient	HygieneArea
SurfaceCategories					
Equipment	0.000	0.031	-0.021	-0.030	0.020
Patient	-0.188	0.120	0.040	-0.025	0.052
NearPatient	-0.562	0.108	0.294	0.162	-0.003
FarPatient	-0.342	0.099	-0.031	0.212	0.061
HygieneArea	0.099	-0.018	-0.038	-0.007	-0.035

For 7x7 matrix

Next_Category	In	Equipment	Patient	NearPatient	FarPatient
HygieneArea	Out				
SurfaceCategories					

In	0.000	0.020	0.000	0.000	-0.020
0.000 0.000					
Equipment	0.000	-0.004	0.029	-0.021	-0.031
0.020 0.007					
Patient	0.000	-0.188	0.120	0.040	-0.025
0.052 0.000					
NearPatient	0.000	-0.562	0.108	0.294	0.162
-0.003 0.000					
FarPatient	0.000	-0.216	0.081	-0.018	0.196
0.051 -0.095					
HygieneArea	-0.002	0.091	-0.017	-0.036	-0.008

-0.035 0.006
Out 0.000 0.000 0.000 0.000 0.000
0.000 0.000

Mean Absolute Error (Train vs Test) for 5x5: 0.1039
Mean Absolute Error (Train vs Test) for 7x7: 0.0522

=== Levenshtein Distance Summary ===

=== Method: (5x5_fixed) ===

Observed chains: 49

Simulated chains: 1000

Total comparisons: 49 x 1000 = 49,000

Simulation time: 0.00 seconds

Calculation time: 0.02 seconds

Absolute Distance:

Mean: 16.13

Median: 15.00

Mode: 14.00

Std: 4.49

Min: 5

Max: 37

Skew: 1.642

Normalised Distance:

Mean: 0.6019

Median: 0.5897

Mode: 0.5000

Std: 0.1258

Min: 0.1765

Max: 1.0000

Skew: 0.296

Normal Fit (Absolute): $\mu = 16.13$, $\sigma = 4.49$

Normal Fit (Normalised): $\mu = 0.6019$, $\sigma = 0.1258$

=== Method: (5x5_fitted) ===

Observed chains: 49

Simulated chains: 1000

Total comparisons: 49 x 1000 = 49,000

Simulation time: 0.00 seconds

Calculation time: 0.02 seconds

Absolute Distance:

Mean: 17.51

Median: 17.00

Mode: 16.00

Std: 6.32

Min: 2

Max: 51

Skew: 0.744

Normalised Distance:

Mean: 0.6211

Median: 0.6129

Mode: 0.6000

Std: 0.1312

Min: 0.1930

Max: 1.0000
Skew: 0.331
Normal Fit (Absolute): $\mu = 17.51$, $\sigma = 6.32$
Normal Fit (Normalised): $\mu = 0.6211$, $\sigma = 0.1312$

=== Method: (7x7_terminating) ===

Observed chains: 49
Simulated chains: 1000
Total comparisons: 49 x 1000 = 49,000
Simulation time: 0.00 seconds
Calculation time: 0.02 seconds

Absolute Distance:

Mean: 20.96
Median: 17.00
Mode: 16.00
Std: 18.06
Min: 0
Max: 242
Skew: 3.929

Normalised Distance:

Mean: 0.5751
Median: 0.5758
Mode: 0.5000
Std: 0.1463
Min: 0.0000
Max: 1.0000
Skew: -0.264

Normal Fit (Absolute): $\mu = 20.96$, $\sigma = 18.06$
Normal Fit (Normalised): $\mu = 0.5751$, $\sigma = 0.1463$

=== Needleman-Wunsch Summary ===

=== Method: (5x5_fixed) ===

Observed chains: 49
Simulated chains: 1000
Total comparisons: 49 x 1000 = 49,000
Calculation time: 26.90 seconds

Absolute Score (Higher is better/more similar):

Mean: -8.82
Median: -9.00
Mode: -9.00
Std: 5.39
Min: -31.00
Max: 9.00
Skew: -0.252

Normalised Score:

Mean: -0.6822
Median: -0.5556
Mode: -1.0000
Std: 0.6430
Min: -4.7500
Max: 0.6000
Skew: -2.779

Normal Fit (Absolute): $\mu = -8.82$, $\sigma = 5.39$

Normal Fit (Normalised): $\mu = -0.6822$, $\sigma = 0.6430$

=== Method: (5x5_fitted) ===

Observed chains: 49

Simulated chains: 1000

Total comparisons: $49 \times 1000 = 49,000$

Calculation time: 27.26 seconds

Absolute Score (Higher is better/more similar):

Mean: -10.35

Median: -10.00

Mode: -9.00

Std: 6.49

Min: -49.00

Max: 15.00

Skew: -0.622

Normalised Score:

Mean: -0.9356

Median: -0.6923

Mode: -1.0000

Std: 0.9410

Min: -13.0000

Max: 0.5385

Skew: -3.141

Normal Fit (Absolute): $\mu = -10.35$, $\sigma = 6.49$

Normal Fit (Normalised): $\mu = -0.9356$, $\sigma = 0.9410$

=== Method: (7x7_terminating) ===

Observed chains: 49

Simulated chains: 1000

Total comparisons: $49 \times 1000 = 49,000$

Calculation time: 29.69 seconds

Absolute Score (Higher is better/more similar):

Mean: -12.79

Median: -9.00

Mode: -5.00

Std: 16.44

Min: -238.00

Max: 14.00

Skew: -4.151

Normalised Score:

Mean: -1.8616

Median: -1.0000

Mode: 0.0000

Std: 2.4146

Min: -59.5000

Max: 1.0000

Skew: -2.850

Normal Fit (Absolute): $\mu = -12.79$, $\sigma = 16.44$

Normal Fit (Normalised): $\mu = -1.8616$, $\sigma = 2.4146$

=== Smith-Waterman Summary ===

=== Method (5x5_fixed) ===

Observed chains: 49

Simulated chains: 1000
Total comparisons: 49 x 1000 = 49,000
Calculation time: 7.95 seconds
Absolute Score (Higher is better/more similar):
Mean: 10.34
Median: 10.00
Mode: 8.00
Std: 4.79
Min: 0.00
Max: 35.00
Skew: 0.485
Normalised Score:
Mean: 0.3325
Median: 0.3182
Mode: 0.5000
Std: 0.1276
Min: 0.0000
Max: 0.9211
Skew: 0.328
Normal Fit (Absolute): $\mu = 10.34$, $\sigma = 4.79$
Normal Fit (Normalised): $\mu = 0.3325$, $\sigma = 0.1276$

=== Method (5x5_fitted) ===
Observed chains: 49
Simulated chains: 1000
Total comparisons: 49 x 1000 = 49,000
Calculation time: 8.03 seconds
Absolute Score (Higher is better/more similar):
Mean: 9.92
Median: 9.00
Mode: 8.00
Std: 5.17
Min: 0.00
Max: 41.00
Skew: 0.700
Normalised Score:
Mean: 0.3556
Median: 0.3421
Mode: 0.5000
Std: 0.1532
Min: 0.0000
Max: 1.0000
Skew: 0.559
Normal Fit (Absolute): $\mu = 9.92$, $\sigma = 5.17$
Normal Fit (Normalised): $\mu = 0.3556$, $\sigma = 0.1532$

=== Method (7x7_terminating) ===
Observed chains: 49
Simulated chains: 1000
Total comparisons: 49 x 1000 = 49,000
Calculation time: 8.66 seconds
Absolute Score (Higher is better/more similar):
Mean: 9.41
Median: 7.00

Mode: 4.00
Std: 5.95
Min: 0.00
Max: 48.00
Skew: 1.413
Normalised Score:
Mean: 0.4833
Median: 0.5000
Mode: 0.6667
Std: 0.1786
Min: 0.0000
Max: 1.0000
Skew: 0.110
Normal Fit (Absolute): $\mu = 9.41, \sigma = 5.95$
Normal Fit (Normalised): $\mu = 0.4833, \sigma = 0.1786$

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SENSITIVITY

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=== Sensitivity Analysis ===
=== 5x5 Matrix Sensitivity ===
Noise Level 0.1%: MAE = 0.0187 ± 0.0037
Noise Level 0.5%: MAE = 0.0189 ± 0.0040
Noise Level 1.0%: MAE = 0.0198 ± 0.0035
Noise Level 2.0%: MAE = 0.0206 ± 0.0036
Noise Level 5.0%: MAE = 0.0302 ± 0.0063
5x5 Model sensitivity: LOW

=== 7x7 Matrix Sensitivity ===
Noise Level 0.1%: MAE = 0.0106 ± 0.0021
Noise Level 0.5%: MAE = 0.0109 ± 0.0023
Noise Level 1.0%: MAE = 0.0119 ± 0.0027
Noise Level 2.0%: MAE = 0.0129 ± 0.0028
Noise Level 5.0%: MAE = 0.0222 ± 0.0042
7x7 Model sensitivity: LOW

=== Sensitivity Comparison ===

Noise Level	5x5 MAE	7x7 MAE	More Robust
0.1%	0.0187	0.0106	7x7
0.5%	0.0189	0.0109	7x7
1.0%	0.0198	0.0119	7x7
2.0%	0.0206	0.0129	7x7
5.0%	0.0302	0.0222	7x7

Overall Assessment:
5x5 matrix: Robust (MAE = 0.0302)
7x7 matrix: Robust (MAE = 0.0222)
Conclusion: Both models show similar sensitivity to noise

=== Bootstrap Analysis ===

Number of samples: 100

5x5 Matrix Variability:

Mean absolute difference from bootstrap mean: 0.0018

Average standard deviation across entries: 0.0207

Most variable transition: (np.int64(2), np.int64(2))

Max standard deviation: 0.0685

7x7 Matrix Variability:

Average standard deviation across entries: 0.0111

Max standard deviation: 0.0685

=== Chi-Squared Tests ===

5x5 Matrix (Observed vs Predicted):

Chi-squared statistic: 124.0096

Degrees of freedom: 23

P-value: 0.000000

Number of cells tested: 24

Result: Difference is highly significant ($p < 0.001$)

7x7 Matrix (Observed vs Predicted):

Chi-squared statistic: 119.3013

Degrees of freedom: 29

P-value: 0.000000

Number of cells tested: 30

Result: Difference is highly significant ($p < 0.001$)

Train vs Test 5x5 Matrix:

Interpretation: Overfitting likely

=== Missing Events Analysis ===

5x5 Matrix - Transitions that never occurred in training data:

Total missing: 1

HygieneArea → NearPatient

Categorisation:

Probably impossible: 0

Rare but possible: 1

HygieneArea → NearPatient

7x7 Matrix - Missing transitions: 19

=== Do Simulations Predict Missing Events? ===

Events that were missing in training but predicted by 5x5 simulations: 0

None - simulations only predict events that were observed in training

Events that were missing in training but predicted by 7x7 simulations: 0

None - simulations only predict events observed in training

=== Summary of Hellinger Distances ===

Method: 5x5_fixed

Care Period Lengths: 0.7720

Equipment: 0.3662

Patient: 0.2879

NearPatient: 0.1726

FarPatient: 0.3085

HygieneArea: 0.5022

Average Hellinger Distance (Category Visits): 0.3275

Method: 5x5_fitted

Care Period Lengths: 0.2468

Equipment: 0.2816

Patient: 0.3270

NearPatient: 0.1691

FarPatient: 0.3579

HygieneArea: 0.4010

Average Hellinger Distance (Category Visits): 0.3073

Method: 7x7_terminating

Care Period Lengths: 0.2927

In: 0.4144

Equipment: 0.2121

Patient: 0.2529

NearPatient: 0.1658

FarPatient: 0.3252

HygieneArea: 0.1963

Out: 0.1763

Average Hellinger Distance (Category Visits): 0.2490